

Lab 1A: String Induction

The following problems ask you to prove some “obvious” claims about recursively-defined string functions. In each case, we want a self-contained, step-by-step induction proof that builds on formal definitions and prior results, *not* on intuition. In particular, your proofs must refer to the formal recursive definitions of string length and string concatenation:

$$|w| := \begin{cases} 0 & \text{if } w = \varepsilon \\ 1 + |x| & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

$$w \bullet z := \begin{cases} z & \text{if } w = \varepsilon \\ a \cdot (x \bullet z) & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

You may freely use the following results:

Lemma 1. $w \bullet \varepsilon = w$ for all strings w .

Lemma 2. $|w \bullet z| = |w| + |z|$ for all strings w and z .

Lemma 3. $(w \bullet y) \bullet z = w \bullet (y \bullet z)$ for all strings w, y , and z .

The **reversal** w^R of a string w is defined recursively as follows

$$w^R := \begin{cases} \varepsilon & \text{if } w = \varepsilon \\ x^R \bullet a & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

For example, $\text{STRESSED}^R = \text{DESSERTS}$ and $\text{WTF374}^R = \text{473FTW}$.

1. Prove that $|w| = |w^R|$ for every string w .

Assume that $|x| = |x^R|$ for every string x shorter than w .

Case 1: Suppose that $w = \varepsilon$.

Then,

$$\begin{aligned} |w^R| &= |\varepsilon^R| \\ &= |\varepsilon| \text{ by definition of reversal} \\ &= |w| \end{aligned}$$

Case 2: $w = ax$ for some symbol a and string x .

Then,

$$\begin{aligned}
|w^R| &= |x^R \bullet a| && \text{by definition of reversal} \\
&= |x^R| + |a| && \text{by definition of **Lemma 2**} \\
&= |x^R| + 1 && \text{by definition of length (since } |a| = |a\varepsilon| = 1 + |\varepsilon| = 1) \\
&= |x| + 1 && \text{by definition of inductive hypothesis} \\
&= |ax| && \text{by definition of length} \\
&= |w|
\end{aligned}$$

2. Prove that $(w \bullet z)^R = z^R \bullet w^R$ for all strings w and z .

Assume that $(x \bullet z)^R = z^R \bullet x^R$ for every string x shorter than w (and every string z)

Case 1: Suppose that $w = \varepsilon$.

Then,

$$\begin{aligned}
(w \bullet z)^R &= (\varepsilon \bullet z)^R \\
&= z^R && \text{by definition of concatenation} \\
&= z^R \bullet \varepsilon && \text{by definition of **Lemma 1**} \\
&= z^R \bullet \varepsilon^R && \text{by definition of reversal} \\
&= z^R \bullet w^R
\end{aligned}$$

Case 2: Suppose that $w = ax$ for some symbol a and string x .

Then,

$$\begin{aligned}
(w \bullet z)^R &= (ax \bullet z)^R \\
&= (a \cdot (x \bullet z))^R && \text{by definition of concatenation} \\
&= (x \bullet z)^R \bullet a && \text{by definition of reversal} \\
&= (z^R \bullet x^R) \bullet a && \text{by definition of inductive hypothesis} \\
&= z^R \bullet (x^R \bullet a) && \text{by definition of **Lemma 3**} \\
&= z^R \bullet w^R && \text{by definition of reversal}
\end{aligned}$$

3. Prove that $(w^R)^R = w$ for every string w

Assume that $(x^R)^R = x$ for every string x shorter than w .

Case 1: Suppose that $w = \varepsilon$.

Then,

$$\begin{aligned}
(w^R)^R &= (\varepsilon^R)^R \\
&= \varepsilon^R && \text{by definition of reversal} \\
&= \varepsilon && \text{by definition of reversal} \\
&= w
\end{aligned}$$

Case 2: Suppose that $w = ax$ for some symbol a and string x

Then,

$$\begin{aligned}
(w^R)^R &= ((ax)^R)^R \\
&= (x^R \bullet a)^R && \text{by definition of reversal} \\
&= a^R \bullet (x^R)^R && \text{by Problem 2} \\
&= a^R \bullet x && \text{by the inductive hypothesis} \\
&= a \bullet x && \text{by definition of reversal } (a = a\varepsilon; a^R = \varepsilon^R \bullet a = \varepsilon \bullet a = a) \\
&= ax && \text{by definition of concatenation} \\
&= w
\end{aligned}$$